HZBRL

Let's spend some quality time so that you won't have to pursue farming...

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Khulna University

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Why are we here???

Figure: what and why...??? [**?**]

Table of contents I

Lets Start: Why Ho Zo Bo Ro Lo (Khichuri) I

- No specific goals
- Includes many things that you should know
- Acadeemia won't teach you
- Will try to practice
- stop me AND ASK QUESTIONS
- I AM HERE TO ASSIST YOU
- Needs TIME! (almost 3-4 hours)
- Attention Please...
- Avoid sufferings in the long run

Roadmap: How to become a/an ???

- https://roadmap.sh/
- https://roadmap.sh/frontend
- https://roadmap.sh/frontend/projects
- https://roadmap.sh/backend
- https://roadmap.sh/backend/projects
- https://roadmap.sh/full-stack

How Things Start (Bootloader & OS): The First Spark I

- **BIOS/UEFI:** The very first code that runs when you press the power button.
- **Bootloader (GRUB/Windows Boot Manager):**
  - Small program that loads the OS into memory.
  - **GRUB:** Common in Linux, allows you to choose between different Kernels or OSs.
- **The Kernel:** Once loaded, it takes over the hardware.

Files & Hierarchies: Text vs Binary I

- **Text Files:** Readable by humans (C++ source, Python scripts, .txt). Encoded in ASCII/UTF-8.
- **Binary Files:** Compiled programs, images, videos. Looks like "garbage" if opened in Notepad.
- **The "Extension" Lie:**
  - Extensions (.exe, .png, .cpp) are just hints for the OS.
  - **Demo:** Rename `script.py` to `script.txt`. It still contains the same data!
  - Changing a `.docx` to `.zip` and opening it (it's actually a collection of XMLs!).

Files & Hierarchies: Directory Hierarchy I

Windows (Drive-based)

- `C:\` (Root)
- `C:\Windows` (System)
- `C:\Users\Name` (Home)
- Backslashes \

Linux (Root-based)

- / (Root)
- `/bin`, `/etc`, `/home`
- Everything is a file!
- Forward slashes /

Avoid Spaces. They are troublesome.

- `"/mnt/someProject/some path"`
- `'/mnt/someProject/some path'`
- `/mnt/someProject/some\ path`

Under the Hood (Compilation & Execution): How Code Actually Runs

- **The Translation Pipeline:**

Stage	Tool	Input	Output
Preprocessing	Preprocessor	.c (Source)	.i (Expanded Source)
Compiling	Compiler	.i	.s (Assembly)
Assembling	Assembler	.s	.o (Object Code)
Linking	Linker	.o + Libs	.exe (Executable)

- **How Compiled Code Works:**

Simple Exeecution Path

SSD (Secondary Memory) $\rightarrow$ RAM (Primary Memory) $\rightarrow$ CPU

- **Analogy:** Python is like a high-level recipe; Assembly is the specific muscle movements the chef makes.

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1. The High-Level Concept (Python)

```python
a = 5
b = 4
c = a + b
print(c) # Simple and readable

```

Listing 1: Python code for sum and print

- **Abstraction:** We don't care about memory addresses.
- **Interpreter:** Python handles the "translation" on the fly.

AMD64 Architectire: Addition + Print (Linux Syscall) I

```
section .bss
    digit resb 2        ; Reserve 2 bytes for the character + newline
section .text
    global _start
_start:
    ; 1. Perform Math
    mov rax, 5
    mov rbx, 4
    add rax, rbx        ; RAX = 9

    ; 2. Convert Integer to ASCII
    ; ASCII '0' is 48. To get '9', we add 48 to the integer 9.
    add rax, 48
    mov [digit], al     ; Store the lower 8 bits (the char) into memory
    mov byte [digit+1], 10 ; Add a newline character (\n)

```

AMD64 Architectire: Addition + Print (Linux Syscall) II

```
    ; 3. Syscall: sys_write (Print to Screen)
    mov rax, 1          ; Syscall number for write
    mov rdi, 1          ; File descriptor (stdout)
    mov rsi, digit      ; Address of our buffer
    mov rdx, 2          ; Number of bytes to print
    syscall

    ; 4. Syscall: sys_exit (Clean Exit)
    mov rax, 60         ; Syscall for exit
    xor rdi, rdi        ; Return code 0
    syscall

```

Listing 2: AMD64 Architecture Assembly Code (what is actually happening inside your laptop)

Intel 8086: Addition + Print (16-bit)

```
MOV AX, 5
MOV BX, 4
ADD AX, BX     ; AX = 9

; --- Printing Logic (Single Character) ---
ADD AX, 48     ; Convert integer 9 to ASCII '9' (48 is '0')
MOV DL, AL     ; Move result to DL (Data Register)
MOV AH, 02h    ; Function 02h of INT 21h is "Display Character"
INT 21h        ; Call DOS interrupt to print

```

Listing 3: Assembly code for Intel-8086 (Technolgy of 1978)

The Final Layer: Assembly vs. Machine Code

Human-Readable (ASM)

```
MOV AX, 5
MOV BX, 4
ADD AX, BX

; Printing Logic
ADD AX, 48
MOV DL, AL
MOV AH, 02h
INT 21h
```

Computer-Readable (Binary)

```
10111000 00000101 00000000
10111011 00000100 00000000
00000001 11011000


00000101 00110000 00000000
10001000 11000010
10110100 00000010
11001101 00100001
```

- Why CPU Architecture matters?
- Why OS Matters?

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- Why OS Matters?

Under the Hood (Compilation & Execution): The Life of a Process (Live Demo Idea) I

- **The Infinite Loop:** Write a code that does `while True` and observe the process.
- **Observation:** Run it and show the CPU spike in Task Manager or `htop` or `btop`.
- **The "Kill" Command:**
  - Show how to find the **PID** (Process ID).
  - Demonstrate killing it via CLI: `kill -9 <PID>` (Linux/Mac) or `taskkill /F /PID <PID>` (Windows).
- **Lesson:** Code isn't just text; once it runs, it's a living process managed by the OS.

The Gateway (CLI & Shells): Intro to the OS & CLI I

- **The OS as a Middleman:** Explain that the CLI is just a way to talk to the OS Kernel without a "pretty" UI.
- **Shell vs. Terminal:**
  - **The Shell:** The logic (Bash, Zsh, PowerShell). Think it like Shell Programming Language.
  - **The Terminal:** The window that displays it.
- **Examples:** Windows: Command Prompt, Powershell; Ubuntu/Linux: Terminal, Konsole, Xfce-terminal, Warp, Kitty, Ghostty, etc.

Navigation 101 I

- **Where am I?:** `pwd` (Linux) or `cd` (Windows - no args). pwd = print working directory
- Directory == Folder
- Default Directory: `C:\Users\myUserName>`
- Every file/folder has its own path (unique)
- **Moving around:**
  - `.` → Current directory.
  - `..` → Parent directory.
  - `cd ..` or `cd ./..` → Go up one level.
  - `cd ../../` or `cd ./../../` → Go up two levels.
  - `myscript.sh` or `./myscript.sh` → Run or execute something *here* (here means current directory).
- **Paths:**
  - **Absolute:** From the root (`/home/user/doc` or `C:\Users\Doc`).
  - **Relative:** From where you are right now (`../images/photo.jpg`).

The Gateway (CLI & Shells): Practice CLI

- **Navigation & View**
  - `dir` → List files/folders. (`dir /p`)
  - `tree` → Graphical folder structure. (`tree /f`)
  - `cd` → Change directory. (`cd Documents`)
  - `cls` → Clear screen. (`cls`)
- **File Management**
  - `mkdir` → Create a new folder. (`mkdir Lab_Work`)
  - `ren` → Rename file/folder. (`ren old.txt new.txt`)
  - `del` → Delete a file. (`del temp.log`)
- **Data & Attributes**
  - `type` → View file content. (`type readme.txt`)
  - `find` → Search for text. (`find "Error" log.txt`)
  - `attrib` → Change attributes (Read-only, Hidden). (`attrib +h private.txt`)

For More: Visit this preseentation (click here)

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The Goal: Running HelloKUCSE Everywhere I

Problem: The OS doesn't know where your executable lives unless you are inside that specific folder.

Windows (CMD)

Phase 1: The Fail

```
cd \
HelloKUCSE.exe
# Output: 'HelloKUCSE.exe'
# is not recognized...
```

Linux (Bash)

Phase 1: The Fail

```
cd ~
HelloKUCSE
# Output: command not found:
# HelloKUCSE
```

The Solution: Setting the PATH

We define a variable KUCSE_HOME and append it to the global PATH.

Windows Syntax

```
set KUCSE_HOME=C:\Users\Path\To\Demo
set PATH=%PATH%;%KUCSE_HOME%
HelloKUCSE.exe  # Success!
HelloKUCSE  # Success!
```

Linux Syntax

```
export KUCSE_HOME="/home/user/demo"
export PATH="$PATH:$KUCSE_HOME"
HelloKUCSE      # Success!
```

The Gateway (CLI & Shells): The "Persistence" Problem I

- **What is an Alias?** A way to map a short or custom keyword to a longer command or executable.
- **Scenario:** We want to type `HelloHello` to run our program `HelloKUCSE.exe`.

The Gateway (CLI & Shells): The "Persistence" Problem II

Windows (CMD)

Using doskey

```
# Syntax:
# doskey alias=command

doskey HelloHello=HelloKUCSE.exe

# To Run:
HelloHello
```

Linux (Bash/Zsh)

Using alias

```
# Syntax:
# alias name='command'

alias HelloHello='./HelloKUCSE'

# To Run:
HelloHello
```

The Gateway (CLI & Shells): The "Persistence" Problem III

- **Session Scope:** By default, these aliases only exist in the *current* terminal window.
- **Making it Permanent:**
  - **Linux:** Add the command to your `~/.bashrc` or `~/.zshrc` file.
  - **Windows:** Requires creating a `.bat` file and linking it via the Registry (AutoRun) or using PowerShell profiles.
- **Common Use Case:** `alias gs='git status'` or `alias ll='ls -la'`.

The Gateway (CLI & Shells): CLI Parameters & Execution I

- **Arguments/Flags:** Programs aren't just names; they take instructions.
  - `ping 8.8.8.8` (Argument: IP)
  - `ping -h` or `ping --help` (Flag: show help)
- **CLI vs. Double-Click:**
  - Double-click: OS opens a "shell-less" window. If a program just prints and exits, the window disappears instantly! (REMINDER: EXERCISE)
  - CLI: The shell stays open, so you can see the `stdout` (output) and `stderr` (errors).

The Gateway (CLI & Shells): How GUI Actually Works I

- **The Event Loop:** GUI programs don't just "run and end"; they "listen."
- *"Is the mouse clicked?"* → *"No"* → *"Wait"* → *"Is the mouse clicked?"*
- **Layers:** Hardware → Kernel → Window Manager (X11/Wayland/DWM) → Toolkit (GTK/Qt/Win32) → Your App.

The Gateway (CLI & Shells): Choosing Your Flavor (Multiple CLIs) I

- **Bash:** The industry standard (Linux/macOS).
- **Zsh:** Bash but "prettier" (Default on modern Macs).
- **Fish:** The "friendly" shell with auto-suggestions.
- **PowerShell:** The powerhouse for Windows (Object-oriented).

C/C++: Deep Dive: Compilers I

- **What is a Compiler?** It translates high-level C++ to Machine Code.
- **The "Big Three":**
  - **GCC (GNU):** The Linux standard.
  - **Clang (LLVM):** Great error messages, used by Apple/Google.
  - **MSVC:** The Windows (Visual Studio) standard.
- **CLI Compilation:**
  - `g++ main.cpp -o myapp`
  - `./myapp`

C/C++: Deep Dive: Let's compile a source code with CLI I

- `gcc hello.c` will output an executable file named `"a.exe"` or `"a"`
- `gcc hello.c -o helloCSEKU` will output an executable file named `"helloCSEKU.exe"` or `"helloCSEKU"`
- `gcc -std=c23 hello.c -o helloCSEKU` will compile in C-2023 language standard/version
- `gcc -S hello.c`
- `objdump -d hello`

C/C++: Deep Dive: Headers & Preprocessors

- **What is a Header (.h/.hpp)?**
  - A promise to the compiler: "I will define this function later, just trust me for now."
  - **The Cherno's Guide:** `https://youtu.be/9RJTQmK0YPI` (Watch this!)
- **#include:** Literally "Copy and Paste" the content of that file here.
- **#define:** A search-and-replace tool. (REMINDER: EXAMPLE)

```
#define PI 3.1415
#define SHOW(x) std::cout << x << std::endl;
```

- **Where are these header files?**
- What does `#include <math.h>` mean ?
  - Linux: `/usr/include/math.h`
  - Windows MinGW: `C:\MinGW\include\`
  - Windows MSYS2: `C:\MSYS2\usr\include\math.h`
- Can I use `#include <./Rifat/MyFevHeader.h>` ?
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- Diffeerence between `#define` and useer defined functions ?

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C/C++: Deep Dive: Communication with Code (Argc/Argv) I

```c
int main(int argc, char* argv[]) {
    // argc = number of arguments
    // argv = the list of strings
    printf("Program Name: %s\n", argv[0]);
    if (argc > 1) {
        printf("First Argument: %s\n", argv[1]);
    }
    return 0;
}

```

Listing 4: How code sees CLI arguments

Lets practice: `https://github.com/KaziRifatMorshed/Quote-of-my-day`

C/C++: Deep Dive: Build Configurations I

- **Debug:**
  - Optimizations are turned off (makes debugging easier).
  - Includes debug symbols (so you can see line numbers and variable names in a debugger).
  - Much larger and slower binary.
  - Does not contain shared libraries with the executable file.
- **Release:**
  - Fully optimized for speed and size (-O2, -O3 flags). (REMINDER: EXERCISE)
  - Debug symbols are stripped out.
  - The compiler may reorder or remove "useless" code to gain performance.
  - Contains shared libraries with the executable file.
- **The Core Difference:** Debug is for the *Developer* (to find bugs); Release is for the *User* (for performance).

C/C++: Deep Dive: Standard Project Structure I

```
Project/
+-- bin/
+-- build/
+-- include/
+-- lib/
+-- src/
+-- tests/
\-- CMakeLists.txt
```

- **src/:** Source files (.cpp, .c).
- **include/:** Header files (.h, .hpp).
- **build/:** Temporary files generated during compilation (Keep this out of Git!).
- **bin/:** The final executable/binary.
- **lib/:** External libraries or dependencies.
- **tests/:** Unit tests to verify your code.

C/C++: Deep Dive: Build Tools (Why CMake?) I

- **The Problem:** Compiling 1000 files manually with g++ is impossible.
- **The Solution: CMake** / **QMake**.
- They don't compile code; they *generate* instructions (Makefiles/Project Files) on how to compile.

CMake Example

1. Create and enter the build directory
mkdir build; cd build
2. Configure the project (Generates Makefiles)
cmake ..
3. Build the project (Compiles the code)
cmake –build .

- EXERCISE: `https://github.com/ttroy50/cmake-examples`

C/C++: Deep Dive: GUI in C (GTK Example) I

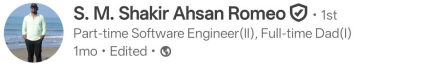
```c
#include <gtk/gtk.h>

int main(int argc, char *argv[]) {
    gtk_init(&argc, &argv);
    GtkWidget *window = gtk_window_new(GTK_WINDOW_TOPLEVEL);
    gtk_window_set_title(GTK_WINDOW(window), "Hello HZBRL!");
    g_signal_connect(window, "destroy", G_CALLBACK(gtk_main_quit),
   NULL);
    gtk_widget_show(window);
    gtk_main();
    return 0;
}

```

Listing 5: Simple GTK Window

C/C++: Deep Dive: GUI in C (GTK Example) II

- **Docs:** `https://docs.gtk.org/`
- Notice how it uses `argc/argv` and an "Event Loop" (`gtk_main()`).

Copilot vs Slave: Myself vs AI I



S. M. Shakir Ahsan Romeo · 1st
Part-time Software Engineer(II), Full-time Dad(I)
1mo • Edited •

Recently, I have been generating more code with AI than I used to before. Specially, the unit tests. For example, yesterday, I wrote a java class with a single method of only 3 lines. Then I told kiro to to generate the unit tests. And it wrote 10 test methods for different scenarios.
I went through them line by line. And thought had I written them myself, I would have written a few less.

What I am trying to tell is even though AI is writing these, I check almost everything before committing. Because the author of the commit is me at the end of the day and when something breaks, nobody will tell "AI broke this", but "Romeo broke this".

Honestly, I have a tendency of owning the code. In principal, I am still an "organic" coder.

What do you think about yourself?

You and 88 others 11 comments

Copilot vs Slave: Myself vs AI II

The "don't push to production on Friday" rule is a long-standing industry meme, but it gained massive mainstream attention due to the CrowdStrike outage on Friday, July 19, 2024. While the rule existed for years to prevent weekend-ruining bugs, the CrowdStrike incident became the ultimate "cautionary tale" because:

The Timing: **CrowdStrike pushed a faulty "content update"** early Friday morning (UTC). The Impact: It caused **approximately 8.5 million Windows machines to crash (Blue Screen of Death), disrupting airlines, banks, and hospitals globally.**

The Irony: It reinforced the meme so perfectly that it's now often cited as the primary reason why companies should avoid Friday deployments.

2024 CrowdStrike-related IT outages

Multiple blue screens of death caused by a faulty software update on baggage carousels at LaGuardia Airport, New York City

[?]

The Engine Room (Fundamentals): Structuring Code I

- **What is an Interpreter?** Explain how Python reads code line-by-line vs. all at once.
- **Compiler vs. Interpreter:** Use a simple analogy (Translating a book beforehand vs. having a live translator at a meeting).
- **Python Bytecode (.pyc):** Python actually compiles to bytecode before interpreting (e.g., `__pycache__`).
- **Built-in Libraries:** What comes "out of the box" (e.g., `math`, `os`, `random`).
- **Packages & External Libraries:** Code written by others (e.g., `requests`, `numpy`).
- **The `import` Statement:**
  - **Import at Top:** Standard practice (readability, dependency checking).
  - **Import inside functions:** When to use it (preventing circular imports or saving memory).

The Ecosystem (Pip & Venv): Virtual Environments (The "Silo" Concept) I

- **What is Pip?** The "App Store" for Python.
- **Requirements.txt:** Crucial for sharing project dependencies with others.
- **Why use them?** Explain "Dependency Hell" (e.g., Project A needs v1.0, Project B needs v2.0).
- **Creating & Activating:**
  - **CLI:** `python -m venv .venv` and `source bin/activate`.
  - **PyCharm UI:** How to point the "Project Interpreter" to a specific environment.
- **Environment Variables & PATH:** Why typing `python` in the terminal sometimes results in "command not found."

Distributions & Specialized Tools: Why pip important ? I

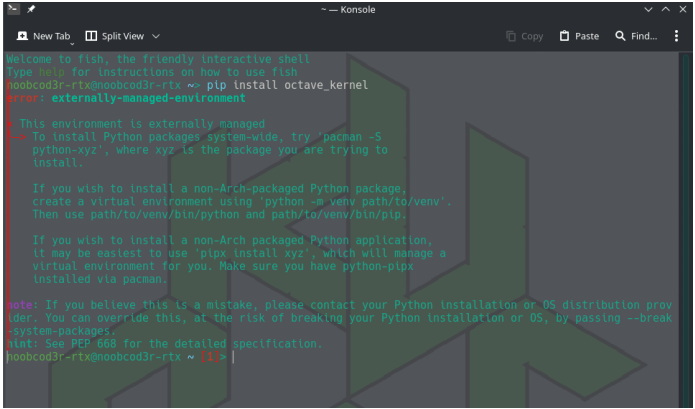


Figure: Error if I try to install a package in base python

Distributions & Specialized Tools: Deployment I

- **Anaconda vs. Miniconda:**
  - **Anaconda:** The "Everything included" (3GB+) version.
  - **Miniconda:** The "DIY" version (Preferred by most pros).
- **Why Conda?** It manages non-python dependencies (like C++ libraries or CUDA for AI) better than Pip.
- **Compiling to Executable:** Tools like `PyInstaller` or `Nuitka`.
- **Bundling:** Explain that this "bundles" the interpreter so the user doesn't need to install Python.

The Data Science & AI Workflow: Hardware for AI (The GPU Question) I

- **Jupyter Notebooks:** Explain the `.ipynb` format—running code in "cells" instead of one big script.
- **Google Colab:** "Jupyter in the Cloud." Explain that it's a VM (Virtual Machine) provided by Google.
- **Why GPU is important?**
  - **CPU:** A few smart people doing complex math (Serial).
  - **GPU:** Thousands of average people doing simple math simultaneously (Parallel).
- **Why are they expensive?** High demand for AI, specialized VRAM, and manufacturing complexity.

Infrastructure (Virtual Terminals & Cloud): Virtual Terminals (The "Azure Experience") I

- **What is it?** It's not a physical computer; it's a terminal window connected to a server thousands of miles away.
- **The "Gist":** You are sending text commands over the internet, the remote OS executes them, and sends the text output back to you.
- **Why it matters:** This is how 99% of professional Python code is deployed (Servers, Cloud Functions, Bots).

Python -¿ code carefully

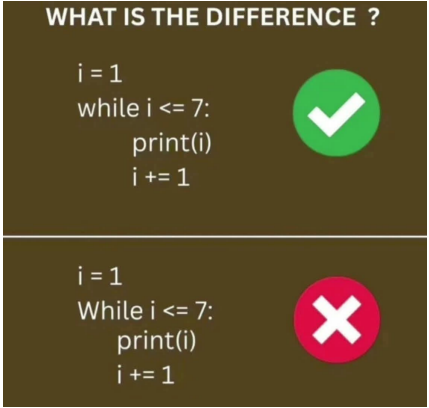


Figure: Find issues [?]

Environment Management (The Practical Side): Python Environment Deep Dive I

- **Import Logic:** Why `import` goes at the top (readability/PEP8) vs. inside a function (lazy loading/avoiding circular dependencies).
- **The Path Variable:** When you type `python`, the CLI looks through a list of folders (the PATH) to find the executable.
- **Package Managers:** `Pip` (Python specific) vs. `Conda` (System-wide library management).

Image Sources I

PlanetF1.
George russell leads leaving pits hungary, 2021.

Smishra1 — Wikipedia contributors.
Crowdstrike bsod at lga, 2024.

Random post from facebook.
Python code diff with two issues, .

Demo Folder